



2025

O24

BOOKLET NO.

107178

Forest Services

Zoology

Time Allowed : Three Hours

Maximum Marks : 200

Medium : English

Type of Paper : Conventional/Descriptive

Question Paper Specific Instructions

Please read each of the following instructions carefully before attempting questions :

1. There are EIGHT questions divided in two Sections, out of which FIVE are to be attempted.
2. Question No. 1 and 5 are compulsory. Out of the remaining questions, THREE are to be attempted choosing at least ONE question from each Section.
3. The number of marks carried by a question/sub-question is indicated against it.
4. Keep in mind the word limit indicated in the question if any.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
8. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
9. Neat sketches may be drawn, wherever required.
10. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.

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P.T.O.



SECTION – A

- Q1.** Write short notes on **any five** from seven. **(8×5=40)**
- Q1.** (a) Describe water vascular system of sea star and comment upon its utility. **8**
- Q1.** (b) Explain in brief parental care in Amphibia. **8**
- Q1.** (c) Write note on green house effect. **8**
- Q1.** (d) Role of pheromones. **8**
- Q1.** (e) Explain honey bee as social insect. Add a note on its economic importance. **8**
- Q1.** (f) Explain in brief importance of correlation in biostatistics. **8**
- Q1.** (g) Explain the principle, components and applications of PCR (Polymerase Chain Reaction). **8**
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- Q2.** (a) With neat, labelled diagram describe modification of mouth parts in insects (Cockroach, Mosquito and Honey bee) **15**
- Q2.** (b) What is pollution ? Explain different types of air pollutants and its effects over living organisms. **15**
- Q2.** (c) Define biological rhythms. Describe various types of biological rhythms with suitable examples. **10**
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- Q3.** (a) Give general features of Aves. How respiratory system helps in flight adaptation in birds ? Illustrate your answer with suitable diagram. **15**
- Q3.** (b) Define communicable disease. Describe causative agent, symptoms, mode of transmission and control measures of AIDS. **15**
- Q3.** (c) Explain in brief chi-square test in biostatistics and give its significance. **10**
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- Q4.** (a) Explain in brief the comparative account of digestive system of Lizard and Pigeon. **15**
- Q4.** (b) Describe life cycle of Mulberry silk moth (Bombyx mori) and give an account of equipment used in sericulture with suitable diagrams. **15**
- Q4.** (c) Explain flame photometry with its principle, working and applications. **10**
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SECTION – B

Q5.	Write short notes on any five from seven.	(8×5=40)
Q5.	(a) Describe mode of transport across Plasma membrane.	8
Q5.	(b) Explain regulation of gene action in Prokaryotes. Write a note on Lac operon.	8
Q5.	(c) Describe Hardy-Weinberg law with application.	8
Q5.	(d) Explain in details vitamins with diagrams.	8
Q5.	(e) Describe Testosterone and functions of Testosterone.	8
Q5.	(f) Explain organogenesis of heart in Chick.	8
Q5.	(g) Describe the process of Mitosis with diagrams in animals.	8
Q6.	(a) Explain in details structure and functions of Lysosomes with diagrams.	15
Q6.	(b) Describe Linkage with its types and suitable examples.	15
Q6.	(c) Describe Kreb's cycle with suitable diagram.	10
Q7.	(a) Describe Renin – Angiotensin system with suitable diagrams.	15
Q7.	(b) What is a fate map ? Describe fate map of gastrula in Chick.	15
Q7.	(c) Describe different stages in evolution of human.	10
Q8.	(a) Describe multiple alleles with ABO Blood group with appropriate crosses and diagrams.	15
Q8.	(b) Describe contraction of Skeletal Muscles by sliding theory with suitable diagrams.	15
Q8.	(c) Describe tetragamete formation in mouse embryo.	10

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